

Letter to André Weil
Bures-sur-Yvette, July 1, 1969

Dear Professor Weil,

Here is the determination, in the case of a function field, of the way in which the constant of the functional equation we had discussed varies. I enclose a copy of a letter from Langlands and some calculations related to it. I shall use only the assertion (L.L.) below.

1. λ -adic representations, after Serre and Grothendieck

Let K be a function field in one variable with field of constants $\mathbb{F}_q$, let X be the set of its places, let E be a number field, let λ be a place of E not dividing q , and let E_λ be the corresponding local field, with ring of integers $\mathcal{O}_\lambda$ and residue field $k_\lambda = \mathcal{O}_\lambda/\lambda$.

A λ -adic representation of K is a continuous representation of $\text{Gal}(\bar{K}/K)$ on a finite-dimensional vector space over E_λ , such that there is a finite subset S of X outside which σ is unramified.

If $v \in X$, we denote by φ_v a corresponding Frobenius substitution, by $F_v = \varphi_v^{-1}$ a geometric Frobenius, by I_v the inertia group of the local Galois group at v , and by σ_v the local representation defined by σ . We put

$$Z_v(\sigma, T) = \det(1 - F_v T, \sigma_v^{I_v})^{-1}$$

(the action of Frobenius on the inertia invariants), and

$$Z(\sigma, T) = \prod_v Z_v(\sigma, T) \in E_\lambda[[T]].$$

Example. A rank-one λ -adic representation of K is the same thing as a homomorphism from the idèle class group of K to $\mathcal{O}_\lambda^*$, continuous for the discrete topology on $\mathcal{O}_\lambda^*$. We denote by $\ell(n)$ the representation corresponding to $x \mapsto |x|^{-n}$, $n \in \mathbb{Z}$, and we put

$$\sigma(n) = \sigma \otimes \ell(n).$$

A theorem of Grothendieck says that if σ is a λ -adic representation of K , then

- a) $Z(\sigma, T)$ is a rational function in $E_\lambda(T)$;
- b) it satisfies a functional equation

$$Z(\sigma^\vee(1), T) = \varepsilon(\sigma, T) Z(\sigma, T),$$

where $\varepsilon(\sigma, T)$ is a monomial. I shall write

$$\varepsilon(\sigma, T) = q^{(1-g)\dim \sigma} \varepsilon_0(\sigma, T), \quad \varepsilon_0(\sigma) = \varepsilon_0(\sigma, 1);$$

this quantity is a λ -adic unit;

- c) if σ also denotes the λ -adic sheaf defined by σ , then

$$\varepsilon(\sigma, T) = \det(-FT, H^*(\sigma)) = \prod_i \det(-FT, H^i(\sigma))^{(-1)^i}.$$

If S is a finite subset of X , put

$$Z_S(\sigma, T) = \prod_{v \notin S} Z_v(\sigma, T)$$

and

$$\begin{aligned} \varepsilon_S(\sigma, T) &= \varepsilon(\sigma, T) \prod_{v \in S} \det(-F_v T, \sigma_v^{I_v})^{-1} \\ &= \text{the monomial asymptotic to } Z_S(\sigma, T) \text{ as } T \rightarrow \infty \\ &= \det(-FT, H^*(\sigma_S)), \end{aligned}$$

where σ_S is the λ -adic sheaf σ , restricted to $X - S$ and extended by 0 at the $v \in S$. We also put

$$\varepsilon_{0,S}(\sigma, T) = q^{(g-1)\dim \sigma} \varepsilon_S(\sigma, T).$$

The conductors are defined by

$$m_v(\sigma_S) = \begin{cases} \dim(\sigma) - \dim(\sigma_v^{I_v}) + \dim \operatorname{Hom}_{I_v}(\operatorname{Sw}_v, \sigma_v), & v \notin S, \\ \dim(\sigma) + \dim \operatorname{Hom}_{I_v}(\operatorname{Sw}_v, \sigma_v), & v \in S, \end{cases}$$

where, for example,

$$\dim \operatorname{Hom}_{I_v}(\operatorname{Sw}_v, \sigma_v) = \dim \operatorname{Hom}_{I_v}(\operatorname{Sw}_v, \sigma_v \bmod \lambda).$$

Here Sw_v is the Swan representation of a sufficiently large quotient of I_v . We omit S from the notation when $S = \emptyset$.

Now let L be a nonempty set of prime places of E , and let S be a finite subset of X . A compatible system of λ -adic representations ($\lambda \in L$), with exceptional set S , is the datum, for each $\lambda \in L$, of a λ -adic representation σ_λ of K , satisfying:

- (i) each σ_λ is unramified outside S ;
- (ii) for $v \notin S$, $Z_v(\sigma_\lambda, T)$ lies in $E(T)$ and is independent of $\lambda \in L$.

We then put

$$Z_v(\sigma, T) = Z_v(\sigma_\lambda, T) \quad (v \notin S), \quad Z_S(\sigma, T) = \prod_{v \notin S} Z_v(\sigma, T),$$

and

$$\varepsilon_S(\sigma, T) = \varepsilon_S(\sigma_\lambda, T).$$

These quantities do not depend on λ ; indeed, ε_S can be defined in terms of Z_S .

2. (L.L.)

We shall admit the following statement. Let σ and ρ be two complex representations of a finite quotient of $\operatorname{Gal}(\bar{K}/K)$. Assume that their conductors are disjoint. With the preceding notation, one then has

$$\begin{aligned} \varepsilon_0(\sigma \otimes \rho, T) &= \varepsilon_0(\sigma, T)^{\dim \rho} \varepsilon_0(\rho, T)^{\dim \sigma} \\ &\quad \times \prod_{v \in X} \det \sigma(F_v)^{-m_v(\rho)} \prod_{v \in X} \det \rho(F_v)^{-m_v(\sigma)}. \end{aligned} \tag{L.L.}$$

3. Variation of the constants

Theorem. Assume that L is infinite, and let σ and ρ be two compatible systems of λ -adic representations ($\lambda \in L$), with disjoint exceptional sets. Then, for every $\lambda \in L$,

$$\begin{aligned} \varepsilon_0(\sigma_\lambda \otimes \rho_\lambda, T) &= \varepsilon_0(\sigma_\lambda, T)^{\dim \rho} \varepsilon_0(\rho_\lambda, T)^{\dim \sigma} \\ &\times \prod_{v \in X} \det \sigma_\lambda(F_v)^{-m_v(\rho_\lambda)} \prod_{v \in X} \det \rho_\lambda(F_v)^{-m_v(\sigma_\lambda)}, \end{aligned} \quad (*)$$

that is, the formula (L.L.) remains valid.

Let S and T be the disjoint exceptional sets for σ and ρ .

Lemma 1. For fixed λ , formula $(*)$ is equivalent to

$$\begin{aligned} \varepsilon_{0, S \cup T}(\sigma_\lambda \otimes \rho_\lambda, T) &= \varepsilon_{0, S}(\sigma_\lambda, T)^{\dim \rho} \varepsilon_{0, T}(\rho_\lambda, T)^{\dim \sigma} \\ &\times \prod_{v \in T} \det \sigma_\lambda(F_v)^{-m_v(\rho_\lambda)} \prod_{v \in S} \det \rho_\lambda(F_v)^{-m_v(\sigma_\lambda)}. \end{aligned} \quad (*_{S, T})$$

If A (respectively B) is the ratio of the two sides of $(*)$ (respectively of $(*)_{S, T}$), the quotient A/B appears as a product of local terms; we shall show that all of them are 1. If $v \in S$, one has to verify

$$\det(-F_v T, \sigma_v^{I_v} \otimes \rho_v) = \det(-F_v T, \sigma_v^{I_v})^{\dim \rho} \det \rho(F_v)^{\dim \sigma_v^{I_v}},$$

which is immediate.

If σ is a representation of $\text{Gal}(\overline{K}/K)$ on a finite-dimensional vector space over k_λ , define $Z(\sigma, T)$ and $Z_S(\sigma, T)$ in $k_\lambda[[T]]$ by the same formulae as above. By reduction modulo λ , it is trivial that if σ is a representation of $\text{Gal}(\overline{K}/K)$ on a finite free $\mathcal{O}_\lambda$ -module, with conductor contained in S , and if σ/λ is the k_λ -representation obtained from it by reduction modulo λ , then:

Lemma 2.

(i) $Z_S(\sigma/\lambda, T)$ is a rational function; one denotes by

$$q^{(g-1)\dim \sigma} \varepsilon_{0, S}(\sigma/\lambda, T)$$

the monomial asymptotic to $Z(\sigma/\lambda, T)$ as $T \rightarrow \infty$.

(ii) If σ and ρ are as above, with conductors contained in disjoint finite subsets S and T of X , the following two conditions are equivalent:

$$\begin{aligned} \varepsilon_{0, S, T}(\sigma \otimes \rho, T) &= \varepsilon_{0, S}(\sigma, T)^{\dim \rho} \varepsilon_{0, T}(\rho, T)^{\dim \sigma} \\ &\times \prod \det \sigma(F_v)^{-m_v(\rho)} \prod \det \rho(F_v)^{-m_v(\sigma)} \pmod{\lambda}, \end{aligned} \quad (a)$$

and

$$\begin{aligned} \varepsilon_{0, S, T}((\sigma/\lambda) \otimes (\rho/\lambda), T) &= \varepsilon_{0, S}(\sigma/\lambda, T)^{\dim \rho} \varepsilon_{0, T}(\rho/\lambda, T)^{\dim \sigma} \\ &\times \prod \det(\sigma/\lambda)(F_v)^{-m_v(\rho)} \prod \det(\rho/\lambda)(F_v)^{-m_v(\sigma)} \quad \text{in } k_\lambda. \end{aligned} \quad (b)$$

We shall deduce from (L.L.) that:

Corollary 3.

(i) If σ is a representation of $\text{Gal}(\overline{K}/K)$ on a finite-dimensional k_λ -vector space, the functions $Z_S(\sigma, T) \in k_\lambda[[T]]$ are rational.

- (ii) If σ and ρ are two such representations, with conductors contained in disjoint finite sets S and T , formula (a) is always valid.

To prove (i), note that, after enlarging E and k_λ if necessary, Brauer's theorem says that σ lifts virtually to a representation of G on a finite-dimensional vector space over E ; one then applies 2.(i) to this representation. Assertion (ii) is proved similarly, from (L.L.) and Lemma 1. It follows that assertion 2.(ii)(a) is always true, and hence that $(*)_{S,T}$ is true modulo λ .

The same technique of reduction to the case of ordinary L -series, or another method, shows that

$$\deg_T(\varepsilon_{0,S}(\sigma_\lambda, T)) = \dim(\sigma)(2 - 2g) - \sum_v m_v(\bar{\sigma}_\lambda, S)$$

(the Ogg–Shafarevich–Grothendieck formula). The first member of this formula is independent of $\lambda \in L$, so that

$$\sum_v m_v(\bar{\sigma}_\lambda, S)$$

is independent of $\lambda \in L$. There is therefore an infinite subset L_1 of L such that the integers $m_v(\bar{\sigma}_\lambda, S)$ and $m_v(\bar{\rho}_\lambda, T)$ are independent of $\lambda \in L_1$. The two sides of $(*)_{S,T}$ are then independent of $\lambda \in L_1$; since they are equal modulo λ for infinitely many λ , they are equal for $\lambda \in L_1$. To complete the proof, it remains only to verify:

Lemma 4. Assume that L is infinite, and let σ be a compatible system of λ -adic representations, with exceptional set S . Then the integers

$$m_v(\bar{\sigma}_\lambda, S)$$

are independent of $\lambda \in L_1$.

We apply the preceding results to σ and to a finite-order character χ of the idèle class group of K , with values in a finite extension of E , and with conductor T disjoint from S . We already know which integers m_v – namely the $m_v(\bar{\sigma}_\lambda, S)$ for $\lambda \in L_1$ – satisfy

$$\begin{aligned} \varepsilon_{0,S \cup T}(\sigma \otimes \chi, T) &= \varepsilon_{0,S}(\sigma, T) \varepsilon_{0,T}(\chi, T)^{\dim \sigma} \\ &\times \prod_{v \in T} \det \sigma(F_v)^{-m_v(\chi)} \prod_{v \in S} \chi(F_v)^{-m_v}. \end{aligned}$$

On the other hand,

$$\begin{aligned} \varepsilon_{0,S \cup T}(\sigma \otimes \chi, T) &= \varepsilon_{0,S}(\sigma, T) \varepsilon_{0,T}(\chi, T)^{\dim \sigma} \\ &\times \prod_{v \in T} \det \sigma(F_v)^{-m_v(\chi)} \prod_{v \in S} \chi(F_v)^{-m_v(\bar{\sigma}_\lambda, S)} \pmod{\lambda}. \end{aligned}$$

Therefore

$$\prod_{v \in S} \chi(F_v)^{m_v(\bar{\sigma}_\lambda, S) - m_v} = 1 \pmod{\lambda}.$$

Since this is true for every χ , it follows that

$$m_v(\bar{\sigma}_\lambda, S) = m_v,$$

which completes the proof.

Yours sincerely,

P. Deligne